

Executive Summary: Stolen Artworks Intelligence Dashboard

Project purpose

This web application is a deployed cultural-heritage analytics dashboard focused on stolen artworks. Its purpose is to help users quickly understand which cases remain missing, which have been recovered, where thefts occurred, which categories dominate the dataset, and how estimated value and recovery status vary over time. The project is designed for the assignment requirement of a dashboard front-end supported by its own database and data pipeline.

Application and architecture

The front-end is built with React, Vite, and Recharts. It contains KPI cards, a theft timeline, status distribution, category distribution, country distribution, a value-versus-year scatter plot, interactive search, status/category filters, and a case-level data table with source links. The back-end uses Netlify Functions to expose API endpoints for summary metrics and case records. Supabase/PostgreSQL is used as the database layer, with a `stolen_artworks` table for cleaned records and an `etl_runs` table for refresh logging.

Data pipeline

The pipeline begins with a curated JSON dataset of high-profile stolen-art cases. The ETL process standardizes titles, artists, countries, categories, dates, recovery status, and estimated values, then calculates a risk score based on missing status, estimated value, artwork category, and time since theft. The transformed records are upserted into Supabase through a server-side script and a Netlify scheduled function. This gives the project a clear refresh mechanism rather than a static-only dashboard.

Insights shown

The dashboard allows users to compare missing, recovered, and unknown cases; identify countries and categories with higher case counts; observe theft-year patterns; and examine high-value records individually. These views support both descriptive storytelling and operational monitoring. The dataset is clearly presented as a teaching dataset compiled from public case information, not as an official law-enforcement registry.

Deployment plan

The project is deployment-ready for Netlify. After the Supabase schema and seed data are added, the Netlify site can be deployed from GitHub with `npm run build` and `dist` as the publish folder. The final demo should use the Netlify URL, satisfying the requirement that the application not be demonstrated from localhost.

Scoring alignment: ETL/database = 4 pts; dashboard visualizations = 3 pts; data refresh mechanism = 2 pts; executive summary and presentation clarity = 2 pts.